From Spectrum to Verdict

What the DEV plugin computes, the physics underneath it, and which of the numbers deserve to be believed.

Why this document exists. The evaluation tab shows roughly twenty numbers. One of them decides the verdict; most of the others are historical, and at least one is inverted with respect to its name. This document writes down what each is, in formulas, so that reading a report does not require reading the source — and puts the physics beside the algebra, because the metric only makes sense with both.

Its companions. *Capture Fidelity* covers the instrument — how a webcam becomes a spectrum. *Light, Pigment and Solvent* covers the sample — what is in the jar and what it does to a photon. This one starts where those finish, at the absorbance curve $A(\lambda)$, and ends at the verdict.

How to read it. Chapter 1 stands alone and contains the claim. Chapter 3 is the physics, chapter 5 the metric that matters; if you read only two, read those. Chapters 6–7 are colour and caveats. Everything historical has been moved to the appendices.

A note on the worked numbers. Two data sets carry every example:

	set	oil	runs	provenance
green	`20270729C`	fresh green	6 re-seats of one fill	`SPEC_capture_quality.md` §16.11.3
brown	`20260731A`	brown	6 re-seats of one fill	§16.13, "series D"

Both are post-rebuild: same instrument, protocol and dilution recipe. They are the most directly comparable green/brown pair on record. Quoted values are the **mean of the six runs**.

CONTENTS

1. The claim

1.1 The winning metric

One number decides the verdict. It is the **Pigment Index** — internally `Pigment ratio · linear baseline`:

$$\text{Pigment Index} = \frac{B_{\text{Soret}}}{B_{\text{Q}}}$$

B_X = mean absorbance in band X, measured above a baseline fitted through two anchor windows (§5).

with B_{Soret} over **440–460 nm** and B_{Q} over **560–580 nm**, both read on the de-spiked absorbance after subtracting a straight line fitted through **520–540** and **620–630 nm**.

> ⭐ **The far anchor moved from 600–630 to 620–630 on 2026-08-03** (`SPEC_capture_quality.md` §16.20). **This document is written throughout on the shipped 620–630 window**; §5a records what the move changed and why, and is the only place the old window's numbers appear as a live comparison. Two consequences worth carrying into every table below: the **scale moved**, so no threshold survives the change untouched; and `r_Q` , the pedestal residual, belongs to its anchor and moved with it (§5a.2).

1.2 What it achieves

	green `20270729C`	brown `20260731A`
Pigment Index	**15.499 ± 0.714**	**10.160 ± 0.197**
at the shipped threshold **T = 12.5**	**+4.20 σ** above	**+11.87 σ** below

The gap is **5.339 = 52.5 % of the brown mean**, against a pooled σ of 0.524 — **Cohen's *d* = 10.20**.

The two classes do not overlap, on any run, in either set.

1.3 What Cohen's *d* means

The figure is used throughout this document, so it is worth defining once. A raw gap between two class means is uninformative on its own: 5.339 units is impressive only if the measurement's own run-to-run scatter is much smaller than that. **Cohen's *d*** divides one by the other:

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_{\text{pooled}}}, \qquad s_{\text{pooled}} = \sqrt{\frac{s_1^{\,2} + s_2^{\,2}}{2}}$$

the difference between the two class means, in units of their pooled standard deviation.

The subscripts 1 and 2 label the two groups being compared — here **1 = green, 2 = brown**:

symbol	meaning	value here
$\bar{x}_1$	the **mean** of group 1 — the average of green's six runs	15.499

symbol	meaning	value here
$\bar{x}_2$	the **mean** of group 2 — the average of brown's six runs	10.160
s_1	the **standard deviation** of group 1 — how much green's runs scatter about their own mean	0.714
s_2	the standard deviation of group 2 — brown's scatter	0.197
s_{pooled}	the two scatters combined into one, as the root-mean-square of s_1 and s_2	0.524

So $d = (15.499 - 10.160) / 0.524 = 5.339 / 0.524 = 10.20$. Note that only the **standard deviations** enter, not the sample sizes: *d* describes how far apart the two *populations* look, not how confident we are about it. Confidence comes from n, and is handled separately in §5.8.

Read out loud, it is **"how many standard deviations apart the two groups are"**. It is dimensionless, so it does not care what units a metric carries, and it is therefore directly comparable **between rival metrics**. That is why every metric in this document is scored with it.

d	conventional reading	overlap of two normal distributions
0.2	small effect	almost complete
0.5	medium	heavy
0.8	**large**	substantial
3	very large	the distributions barely touch
10.20	**our Pigment Index**	**none observed**

Our *d* = 10.20 means the class means sit ten pooled standard deviations apart — which is why no run of either set comes near the other's range.

⚠ Three reading notes. With n = 6 per class the *value* of *d* is loosely estimated, even though its sign and rough size are not in doubt; §5.8 gives the interval that actually matters. A **negative** *d* in this document means the ordering is **inverted** — brown reading higher than green. And *d* comes in more than one recipe: **Appendix B** gives both pooled-SD conventions, the small-sample correction, and which is used where.

The same idea outside statistics

The identical construction is standard in several other fields under other names — worth knowing, because a reader from any of them already understands the figure. The neutral, field-independent term for the whole family is the **standardised mean difference (SMD)**; Cohen's *d* is one recipe within it (Appendix B).

name	field	relation
d′ ("d-prime")	signal detection, psychophysics	the same quantity; the sensitivity of a detector
Fisher's discriminant ratio	pattern recognition	$J = d^2/2$ — here 60.9
Z-score, "sigma level"	process control, Six Sigma	distance from a *limit*, not between two groups
peak resolution R_s	chromatography	separation ÷ typical width — the same shape of idea

The Z-score row is already in use here without the label: the **"+4.66 σ above T"** and **"+9.88 σ below T"** of §1.2 are Z-scores — one group measured against a threshold, rather than two groups against each other.

1.4 Three properties that make the claim worth making

> **1. It is dilution-invariant *by construction*, not by luck.** Every step from the curve to the two band means is linear and homogeneous, so concentration and path length cancel exactly in the quotient. This is proved, not asserted (§5.6). It means the recipe can change without recalibrating the verdict.
>
> **2. It is a ratio of two chemical SPECIES, not a measure of "how much pigment".** What it tracks is intact protochlorophyll against its magnesium-free degradation product protopheophytin. Across our two classes the *total* Q-region absorbance is identical (0.2300 vs 0.2251) while the shape differs at d = 10.3 — nothing is missing, something has been **converted** (§3.3).
>
> **3. It survives the instrument.** Lamp brightness, camera gain, grating efficiency and exposure all scale the absorbance uniformly and divide straight out. A €30 webcam can carry it.

⇒ **Green-versus-brown discrimination works.** That is the claim, and §5.8 is where it is defended.

1.5 What the claim is *not*

⚠ **Precision is not correctness.** d = 10.20 says the instrument reliably distinguishes *these two bottles*. Whether **T = 12.5** is the right place to cut the population of real pumpkin oils is a separate, still-unvalidated question requiring reference oils with independent ground truth (`SPEC_capture_quality.md` §16.10.11a). A precise instrument reading a wrong threshold is confidently wrong.

⚠ **These are re-seat numbers.** Both sets are six re-seats of *one fill*, so they exclude sample preparation entirely. Fill-to-fill scatter remains unmeasured for brown (`SPEC_capability_proof.md` §11.4f B).

1.6 The chain, in one line

$$2\ captures \Rightarrow R(\lambda),\ S(\lambda) \Rightarrow A(\lambda) \Rightarrow \text{despike} \Rightarrow 4\ band\ means \Rightarrow the\ index$$

2. From two captures to an absorbance curve

2.1 The burst mean

Each capture is a burst of **150 frames** (`DevSpectralPlugin.FRAMES`). Frames whose overall brightness is an outlier are rejected, then each wavelength bin is averaged with a sigma-clipped mean:

$$R(\lambda) = \mathrm{mean}_{f \in F_{kept}}\, r_f(\lambda) \qquad S(\lambda) = \mathrm{mean}_{f \in F_{kept}}\, s_f(\lambda)$$

F_{kept} excludes frames rejected by the per-frame MAD test (SPEC_capture_quality.md §14.8 C1).

The reference is pure isopropanol in the same jar, measured in the same session. Dividing by it removes the lamp's own spectrum — the single most important idea in the instrument.

2.2 Transmittance, with a floor guard

$$T(\lambda) = \frac{S(\lambda)}{R(\lambda)} \qquad \textit{for } R(\lambda) > f \cdot \max_\lambda R(\lambda), \qquad f = 6.31 \times 10^{-5}$$

Where the reference is at or below that fraction of its own peak the wavelength is **masked out entirely** rather than divided; below the floor S/R amplifies noise without bound. The constant looks odd because it lives in linear light — it is the historical "1 % of peak" expressed after gamma decoding, $0.01^{2.2}$ (`SPEC_capture_quality.md` §17).

2.3 Absorbance

$$A(\lambda) = -\log_{10} T(\lambda) = -\log_{10} \frac{S(\lambda)}{R(\lambda)}$$

defined only where $T(\lambda) > 0$; other wavelengths carry no value at all.

This is the **Beer–Lambert** quantity, $A = \varepsilon\, c\, l$ — linear in concentration, which is what makes ratios of it behave (*Light, Pigment and Solvent* §2.2).

2.4 De-spiking

Every metric except the "raw" twin rows is computed on the **de-spiked** curve — a moving median with a 7-point kernel:

$$A_d(\lambda_i) = \mathrm{median}\{A(\lambda_{i-3}), \ldots, A(\lambda_{i+3})\}$$

A median *rejects* isolated outliers where a smoothing filter would average them in. The grid spacing is 0.146 nm, so the kernel spans ≈1 nm — narrow enough to leave a 20–30 nm pigment band untouched, wide enough to kill single hot pixels.

⚠ **It does not remove the two named instrument artifacts.** Two narrow features near **473 nm** (FWHM 1.0 nm) and **607 nm** (FWHM 2.7 nm) are wider than the kernel and survive it. Both are **lamp emission lines** that fail to cancel in S/R (§5.9). **Since the far anchor moved to 620–630, both now sit outside every measurement window** and both are harmless to the index. On the old 600–630 anchor the 607 nm line lay *inside* it — one of the reasons the window moved (§5a.1).

3. The physics the metric rests on

This chapter is the short form. The full chemistry — miscibility, turbidity, the vessel — is in Light, Pigment and Solvent, whose §3 this compresses.

3.1 The pigment is protochlorophyll, not chlorophyll

Fruhwirth & Hermetter's review of the Styrian oil pumpkin identifies the oil's colourants explicitly: *"various tetrapyrrol-type compounds like **protochlorophyll (a and b)** and **protopheophytin (a and b)**, the latter being a protochlorophyll lacking the magnesium ion"* [1].

Protochlorophyll is chlorophyll's biosynthetic **precursor**. It lacks chlorophyll's ring-D reduction, so it is a **porphyrin** where chlorophyll is a **chlorin** — and that single bond moves the red absorption band by ~40 nm:

	Soret	red (Qy) band	fluorescence
chlorophyll *a (the textbook default — **not** ours)*	≈ 430 nm	≈ 662–665 nm	≈ 668–675 nm
protochlorophyll(ide) *a* *(ours)*	**≈ 432–440 nm**	**≈ 623–626 nm**	**≈ 630–636 nm**

Protochlorophyllide's Qy is measured at **623 nm** in 80 % acetone and **626 nm** in methanol [2], and the oil's own fluorescence maximum of **635 nm** [1] corroborates it — a fluorescing molecule emits ~10 nm to the red of the transition it absorbs on, putting the absorber at ~625 nm and nowhere near 662.

This matters because our capture window ends at 630 nm. The red band is therefore *at the edge of what we can see*, not beyond it — which is the whole reason the far window behaves as it does (§3.4), and, once that was understood, the reason the anchor was narrowed onto **620–630** so as to sit on the band rather than straddle its foot (§5a.1).

3.2 Soret and Q — where the bands come from

All porphyrin-type spectra share one two-part shape, explained by **Gouterman's four-orbital model** [3]: the visible spectrum is governed by four frontier orbitals, two nearly-degenerate occupied and two nearly-degenerate empty.

band	transition	character
Soret (also **B**)	$S_0 \rightarrow S_2$	**very strong**, blue, ~430–440 nm
Q	$S_0 \rightarrow S_1$	**weak** — 1/5 to 1/10 of the Soret — yellow-green to red

In a metalloporphyrin the magnesium sits on a four-fold symmetry axis (D_{4h}). The two Q transitions are then **degenerate** — equal energy, polarised at right angles in the plane of the ring — and what appears is one origin band **Q(0,0)** (also α) plus a vibronic satellite **Q(1,0)** (β).

> **"Degenerate" carries no sense of decay.** It means two states happen to share an energy. Symmetry is what enforces the sharing, so lowering the symmetry **lifts** the degeneracy and the states separate.

3.3 What roasting does — and why it rearranges rather than removes

Heat and acid **strip the central magnesium out of the ring**, replacing it with two protons. The product is **protopheophytin** (*pheo-*, Greek *phaiós*, "dusky"). It is not only roasting: in pumpkin seeds protopheophytins accumulate as a **storage** degradation product, reported at **1.1–35.5 %** of the protochlorophylls [4].

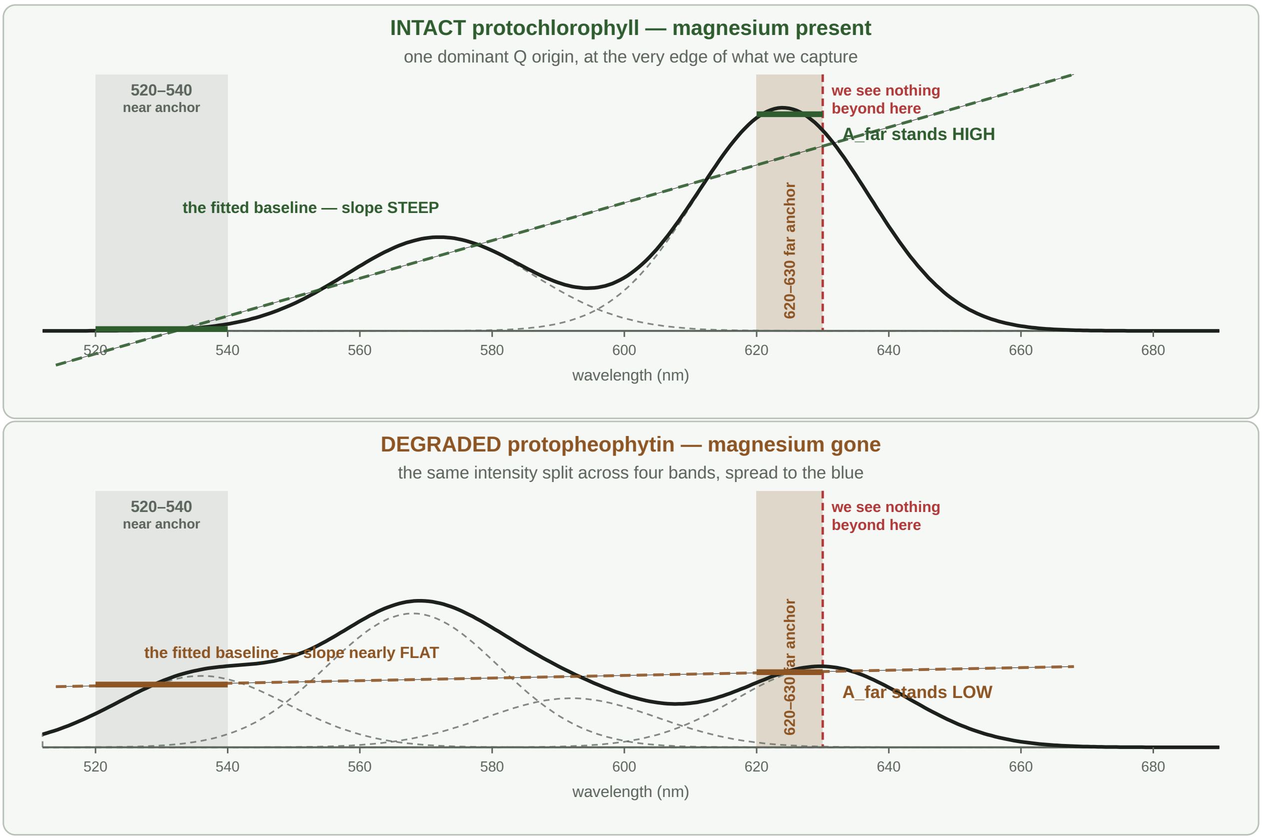
Spectroscopically the two protons sit on **one** axis, so the symmetry drops $\mathbf{D_{4h}} \rightarrow \mathbf{D_{2h}}$, the degeneracy is lifted, and the two Q transitions separate into distinct **Qx** and **Qy** bands. Each keeps its own vibronic satellite, so a metal-free ring shows **four** Q bands, numbered **I–IV from the longest wavelength**: I = Qy(0,0), II = Qy(1,0), III = Qx(0,0), IV = Qx(1,0).

⭐ **And the redistribution has a known direction.** Free-base porphyrin spectra are classified into four types by Q-band intensity ordering — *etio* (IV > III > II > I), *rhodo* (III > IV > II > I), *oxo-rhodo* and *phyllo* [5]. **Band I, the longest-wavelength band, is the weakest in every one of them.** So a pigment whose long-λ band *dominates* while metallated finds it demoted to weakest-of-four once the magnesium goes. Intensity moves toward the blue. *(Protopheophytin carries the ring-E carbonyl, a "rhodofying" group, so* rhodo *is its expected type — band I still weakest.)*

⇒ **The verdict is a ratio of two chemical species**, intact against degraded — not a measure of how much pigment is present.

3.4 ⭐ Why this appears as a change of SLOPE

Why redistributing the Q intensity LOWERS the 620–630 anchor — and FLATTENS the baseline through it



The far window sits on a FLANK, not on a peak, so its height reports the height of the nearest band — not the total pigment.

Measured: A(620–630) − A(520–540) = 0.0804 green vs 0.0488 brown (1.65×), over the 95.0 nm between the window centroids — 8.46 vs 5.14 × 10^{-4} A/nm, which is §5.3's fitted 8.56 / 5.22 to within 1 %. Schematic: pigment bands only, no turbidity pedestal, so its own contrast (17×) is far larger than the measured 1.65×.

⚠ The NEAR anchor is turbidity-dominated in reality, and the two classes nearly coincide there (0.1231 green vs 0.1038 brown), so the real pair of lines pivots about a common left foot. This model has no pedestal, so its left feet differ — read the RIGHT ends against each other.

Figure 1 — the mechanism, and the whole of what the baseline does. **Top:** with the magnesium intact, one dominant Q origin sits at the very edge of our capture window, so the far red — and with it the **620–630 anchor** — stands **high**, and the fitted line through the two anchors is **steep**. **Bottom:** once the magnesium is gone that intensity is spread across weaker bands further blue; nothing tall is left near 630, the anchor drops, and the same line goes **nearly flat**. Total area is similar; only its distribution differs. **The two solid bars are the only numbers the metric takes from the curve** — one mean per anchor window — and the dashed line through them is what gets subtracted before Soret and Q are read. ⚠ Schematic: it models the pigment bands only, with no turbidity pedestal, so its contrast is far larger than the measured 1.65×, and its two near anchors differ where the real ones nearly coincide — **compare the right-hand ends.**

> ⚠ **"Slope" means the FITTED BASELINE's slope, which lives *between* the two windows.** It is easy to read it as the slope *inside* the far window, and the metric never sees that: `linearBaselineCorrected` reduces each anchor to a **single number — its mean absorbance** — and fits a line through the two means. A 10 nm window collapses to one value, so its internal shape is averaged away before the fit begins.
>
> What the far anchor contributes is therefore its **height**, and the slope follows from the two heights by arithmetic — the table below.

	`A_near` 520–540	`A_far` 620–630	difference	÷ 95.0 nm = slope
green	0.1231	0.2035	**0.0804**	8.46 × 10^{-4} A/nm

	`A_near` 520–540	`A_far` 620–630	difference	÷ 95.0 nm = slope
brown	0.1038	0.1526	**0.0488**	5.14×10^{-4} A/nm

— which reproduces §5.3's fitted **8.56 / 5.22 × 10^{-4}** to within 1 %, the remainder being the anchor windows' *internal* slope that the least-squares fit does see and the two-centroid chord does not (§5.5).

> ⚠ **An unresolved tension, visible only at this zoom.** The sourced Qy is **623–626 nm** (`KB_spectroscopy_physics.md` §4.1), which would put the band's *maximum* inside 620–630 — yet the measured green absorbance **rises monotonically** across the window (+0.0766 A over 10 nm) rather than peaking and falling. Either the band shifts red in isopropanol relative to the acetone and methanol values, or the lamp's red collapse biases it: the reference reads 39 DN at 620–630 against 130 at 530 (§16.12.11 B), and a dim reference inflates absorbance. **Both are testable and neither has been tested.** It matters because the shipped anchor sits exactly there — §16.11.17's timed series would see it for free, as would any run that logs the reference level per bin.

Our far window is **narrow and sits on a flank, not on a peak**. The slope across such a window reports **the height of the nearest peak**, not the total pigment. A tall band whose edge crosses the window gives a steep rise; move that intensity 30–50 nm blue and split it, and the window is left on flat ground even though just as much light is absorbed overall.

That makes the far window an unusually **specific** probe: it responds to *whether the intact, symmetric pigment is still there*, and is comparatively blind to how much total pigment the oil contains.

3.5 The far window is a measuring band — measured, not assumed

The far window was introduced as an "oil-quiet" anchor. It is not quiet. Across 37 runs, six fills and two sessions under an identical lamp, the *rise* $A(620\!-\!630) - A(600\!-\!610)$ is **0.0535 for green against 0.0159 for brown — a 5.1 σ separation** (`SPEC_capture_quality.md` §16.12.12). The reference sits at 35–39 DN in both classes, so the lamp is in the same state: an instrument artifact cannot know which oil is in the jar.

And it is load-bearing. Sweeping the far window's right edge inward, Cohen's *d* falls **2.88** → **0.94** and the classes overlap outright by 600–610 nm (§16.12.13). The contamination *carries* the discrimination.

> ⭐ **Both measurements above were made across 600–630**, the anchor of the day, and the shipped window is its red third. They are quoted unchanged because the direction they establish is what motivated the narrowing: the further red the window reaches, the more of the intact pigment it sees. §4.2 shows the shipped 620–630 window separating the classes by **33 %** — a wider class difference than the 30 nm window it replaced, from a third as many points.

> ⛔ **Changing these windows is a metric redesign, not a tidy-up**, and it is gated on post-rebuild data (`SPEC_capture_quality.md` §16.12.16).

3.6 The difference is speciation, not concentration — and the test has no free parameters

Under simple Beer–Lambert with one absorbing species, brown = *k* × green: the curves differ by **one scale factor**, so any ratio of two features taken *inside* the Q region is class-independent. A class difference in such a ratio refutes pure scaling outright.

	green	brown	*d*
`A_Q` 560–580	0.2300 ± 0.0173	0.2251 ± 0.0199	**0.26** — equal
Q amplitude (572 − 550)	0.1275	**0.1495**	**−3.16** — brown HIGHER
rise ÷ Q-amplitude	**0.4274 ± 0.039**	**0.0800 ± 0.028**	**10.26**

Pure scaling predicts *d* = 0 on the last row. Measured *d* = 10.26, a factor of 5.3. Brown is not pigment-poor — `A_Q` is equal and its 572 feature is *stronger*. Normalising each class by its own Q amplitude shows where the intensity went: **out of 615–630 (−0.193) and into 580–615 (+0.07 to +0.11)**, exactly the direction §3.3's band-I rule predicts. *(`diagnostics/qband_shape.py`; `SPEC_capture_quality.md` §16.13.9.)*

3.7 ⚠ Why we cannot simply look at the two-versus-four bands

The natural objection is that a better spectrometer would settle all of this. **It would not:**

grid spacing	0.146 nm/bin, 1305 bins over 440–630
narrowest feature resolved *(473 nm lamp line)*	**FWHM 1.0 nm**
second lamp line *(607 nm)*	**FWHM 2.7 nm**
Q bands to be separated	**20–30 nm wide**

The rig **out-resolves the target by 10–20×**. Four other things stand in the way: **window truncation** at 630 nm (the dominant limit); the two species **always coexisting**, so no pure spectrum of either is ever presented; **20–30 nm intrinsic linewidths** merging the four free-base bands into shoulders; and the **turbidity pedestal** suppressing contrast.

⇒ The remedy is a wider window, not a finer instrument — an argument that redirects an entire class of "buy better hardware" thinking toward a much cheaper change.

4. The windows and the band means

Everything downstream is built from four wavelength windows — two pigment bands and two baseline anchors.

symbol	window	what it sits on	source
A_{Soret}	**440–460 nm**	red flank of the **Soret (B)** band, ~432 nm	`PB_SORET_BAND`
A_{Q}	**560–580 nm**	a band in the **Q region** — assignment open, §7.4	`PB_Q_BAND`
A_{near}	**520–540 nm**	between bands — the quieter anchor	`PB_BASELINE_WINDOWS[0]`
A_{far}	**620–630 nm**	**on** the **Qy(0,0)** band, ~623–626 nm — a *measuring* band (§3.5)	`PB_BASELINE_WINDOWS[1]`

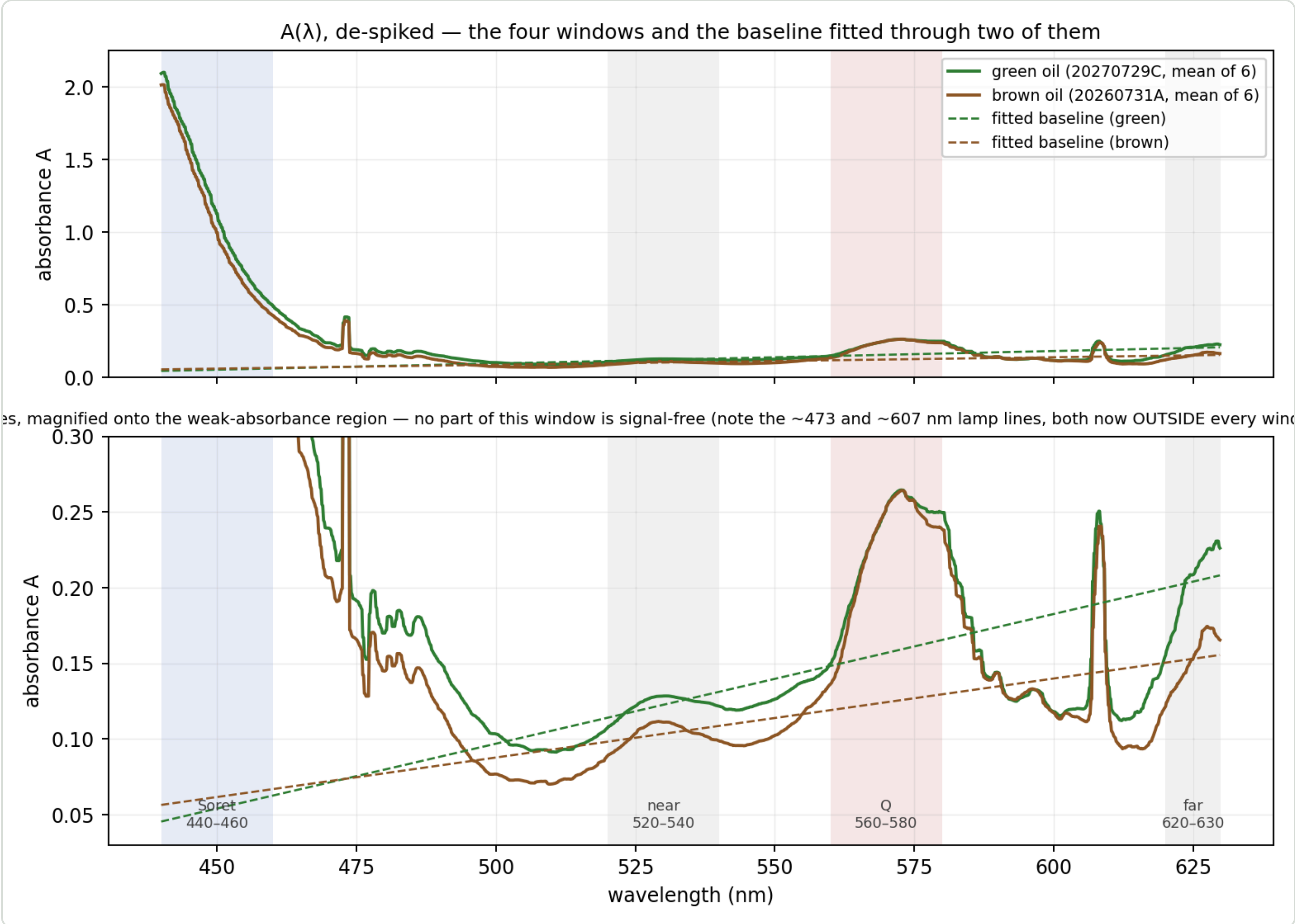


Figure 2 — the de-spiked absorbance of both classes, with the four windows shaded and each curve's own fitted baseline drawn dashed. The lower panel magnifies the weak-absorbance region — **no part of this window is signal-free** (§3.5). Note how little separates the two curves inside the Q window, and how much separates them inside the far window.

4.1 Windows and band means — the notation used from here on

A **window** is a wavelength interval. Write W_{X} for the *set of measured grid points* that fall inside window X:

$$W_X = \{\lambda : lo_X \le \lambda \le hi_X\}$$

read: "all wavelengths λ between the window's lower and upper edge". $|W_X|$ is how many there are.

So W_{Soret} is the 138 grid points between 440 and 460 nm, W_{far} the 71 points between 620 and 630 nm, and so on. The **band mean** is then simply their average:

$$A_X = \frac{1}{|W_X|} \Sigma_{\lambda \in W_X} A_d(\lambda)$$

the sum of the de-spiked absorbance over the window's points, divided by their number.

A plain unweighted mean of every surviving grid point — **not** an integral. For two 20 nm bands the choice makes no difference to the ratio, and a mean keeps the same unit as the other band readings, where an integral would inject a bandwidth factor into the unequal-width comparisons (`SPEC_pumpkin_peak_ratio_eval.md` §9).

4.2 Geometry and measured values

window	points	centroid $\bar{\lambda}$	green	brown	CV green	CV brown
Soret 440–460	138	449.98 nm	1.1864	1.0855	4.04 %	3.58 %
Q 560–580	137	570.02 nm	0.2300	0.2251	7.52 %	8.86 %
near 520–540	135	529.93 nm	0.1231	0.1038	8.89 %	13.31 %
far 620–630	**71**	**624.96 nm**	**0.2035**	**0.1526**	12.99 %	14.54 %

The centroids matter in §5.5. **Read the A_Q row**: 0.2300 against 0.2251, a 2 % difference against 7–9 % scatter. On its own the Q band carries no usable class information (*d* = 0.26).

Now read the A_{far} row against it. 0.2035 against 0.1526 — the two classes differ by **33 %** in the window that was introduced as a *baseline anchor*. That is the largest class difference of any window in the table, and it is why §3.5 calls the far window a measuring band. The 620–630 window is only 10 nm wide and holds a third as many points as its 600–630 predecessor, yet it separates the classes better, because it stands **on** the Qy band instead of straddling its foot (§5a.1).

5. ⭐ The Pigment Index — the metric that works

5.1 What problem the baseline solves

Re-seating the jar tilts the beam slightly, and that enters the absorbance curve as **an offset and a slope together** — the whole curve lifts *and* rotates. A constant subtraction removes the offset; standard normal variate removes offset and scale; **neither removes a slope.** A straight line fitted through two separated anchor windows removes both (`SPEC_capture_quality.md` §16.10.2).

This construction is not new and not ours. It is the **Morton–Stubbs correction for irrelevant absorption**, introduced by Morton and Stubbs in 1946 for the spectrophotometric assay of vitamin A in liver oils, and still in pharmacopoeial use today — in its three-point **Allen** form it is the standard correction for serum haemoglobin against interference from bilirubin and turbidity. *Irrelevant absorption* is the pharmacopoeial term for any absorbance that does not come from the analyte: impurities, degradation products, or, as here, a scattering pedestal. The method asks two things of the measurement: that this irrelevant absorption is **linear across the analytical window**, and that the anchors it is fitted from lie **outside the analyte's own absorption**, so that what they measure is background and nothing else.

Naming it matters, but not as a certificate. Morton–Stubbs rests on two conditions, and this instrument satisfies neither of them.

The first is that the irrelevant absorption is **linear** across the window. A scattering pedestal is convex, so a straight chord over-subtracts in the middle; the size of that over-subtraction is the residual r_Q to which `DOC_pedestal_correction.md` is devoted. It is worth about a quarter of the Q band at working strength.

The second is that the flanking anchors contain **none of the substance being measured**. Ours plainly do. The red anchor at 620–630 nm sits on the pigment's own Qy flank, and comparing it against the scattering law that should govern a true background — the pedestal at 625 nm can be at most about half its value at 530 nm for small particles, or three-quarters for large ones — puts between **half and four-fifths of that anchor's absorbance down to pigment rather than background**. By the method's own standard it is not a correction window at all.

Neither deviation is an oversight, and the second is deliberate. Removing the Qy flank from the anchor has been tried, and the oil classes then overlap: the contamination is carrying the speciation signal the metric exists to detect. Nor does it break the dilution invariance derived later in this chapter, because anchor pigment scales with concentration exactly as the bands do and divides out of the ratio.

What it does cost is comparability. Because a concentration-proportional amount is subtracted from both bands before the ratio is taken, M_∞ is not the pigment's Soret-to-Q ratio in any absolute sense — it is that ratio as seen through this window, on this instrument. That is the deeper reason the threshold is tied to this rig and this recipe rather than to the molecule.

So the honest statement is narrower than "we use the standard method". We use its construction, knowingly outside its stated conditions, and the two departures are quantified above. Naming it earns its place by supplying those conditions as a checklist — which is how the departures came to be measured at all.

5.2 The definition

Take every grid point inside **either anchor window** — W_{near} (520–540 nm, 135 points) and W_{far} (620–630 nm, 71 points), in the notation of §4.1 — and fit **one straight line** through all 206 of them at once.

The line is $A = m\lambda + c$, with $\boldsymbol{m}$ **its slope** in absorbance per nanometre and $\boldsymbol{c}$ **its intercept** — the value the line would take at $\lambda = 0$ nm. The intercept has no physical meaning on its own; it is simply the second number needed to pin a line down once the slope is chosen. What matters is the line's *value across our window*, which §5.3 tabulates.

The fit is **weighted least squares**, with each of the two windows carrying **equal total weight** regardless of how many points it contains. *(What that means and why squares: Appendix A.)*

Subtract the fitted line from the **whole** curve, re-read the two pigment bands on the corrected curve, and divide:

$$B(\lambda) = A_{\mathrm{d}}(\lambda) - (m\lambda + c)$$

$$B_{\mathrm{Soret}} = \frac{1}{|W_{\mathrm{Soret}}|} \Sigma_{\lambda \in \mathrm{W}_{\mathrm{Soret}}} B(\lambda) \qquad B_{\mathrm{Q}} = \frac{1}{|W_{\mathrm{Q}}|} \Sigma_{\lambda \in \mathrm{W}_{\mathrm{Q}}} B(\lambda)$$

$$\text{Pigment Index} = \frac{B_{\mathrm{Soret}}}{\max(B_{\mathrm{Q}},\ \varepsilon)}, \qquad \varepsilon = 10^{-3}$$

the epsilon is a division guard, never reached in practice.

5.3 What the correction does to each band

Measured fits, mean over each set:

	slope $\boldsymbol{m}$	intercept $\boldsymbol{c}$	baseline under Soret	baseline under Q
green	$+8.563 \times 10^{-4}$ A/nm	−0.3312 A	0.0541	**0.1569**
brown	$+5.216 \times 10^{-4}$ A/nm	−0.1730 A	0.0618	**0.1244**

$$B_{\mathrm{Soret}} = 1.1864 - 0.0541 = 1.1323 \qquad B_{\mathrm{Q}} = 0.2300 - 0.1569 = 0.0731 \qquad (\text{green})$$

	$A_{\mathrm{Soret}} \to B_{\mathrm{Soret}}$	$A_{\mathrm{Q}} \to B_{\mathrm{Q}}$
green	1.1864 → 1.1323 *(−5 %)*	0.2300 → **0.0731** *(−68 %)*
brown	1.0855 → 1.0237 *(−6 %)*	0.2251 → **0.1008** *(−55 %)*

The Soret barely notices; the Q band loses more than half its value. That asymmetry follows from geometry — the baseline is a small fraction of a large absorbance and a large fraction of a small one — and it is the whole story of §5.4.

The fitted line is now roughly twice as steep as it was on the 600–630 anchor (+8.56 against +4.81 × 10^{-4} A/nm for green). Nothing about the oil changed: the far anchor moved 10 nm redder and 33 % higher up the Qy flank, so a line pinned through the same near window has to climb harder to reach it. Both lines cross at ≈525 nm, inside the near anchor, which is where two lines fitted through the same window must meet.

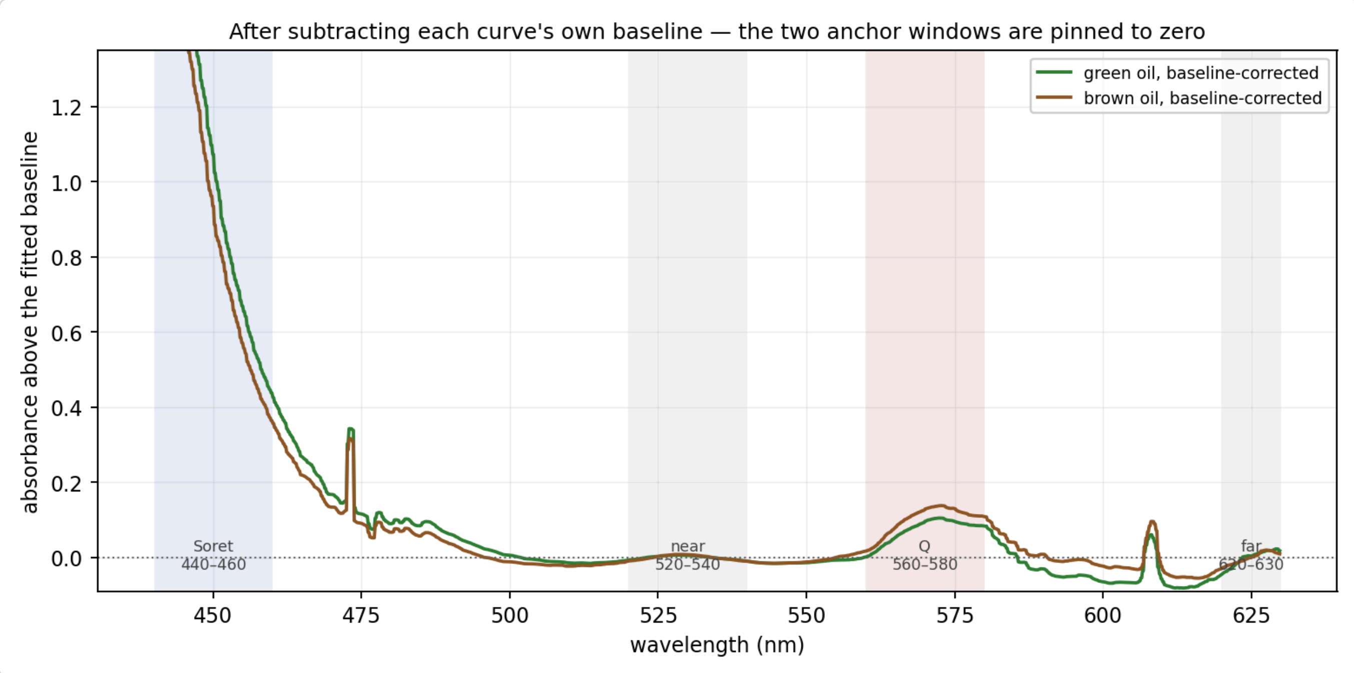


Figure 3 — the same two curves after each has had its own baseline subtracted. The two anchor windows are pinned to zero by construction; everything else is now measured relative to them.

5.3a What the asymmetry costs — the error budget it implies

The table above is usually read as a statement about signal. It is equally a statement about error, and in that reading it is much sharper. If the baseline lands in the wrong place by some small amount, the two bands do not suffer equally: the same absolute error is divided by 1.13 at the Soret and by 0.073 at the Q band.

a 0.001 A baseline error is worth	on the metric
at the Q band	**1.4 %**
at the Soret	0.09 %

The metric is therefore about **fifteen times** more sensitive to where the baseline lands under the Q band than under the Soret — the ratio is simply $B_{\mathrm{Soret}}/B_{\mathrm{Q}}$, so it is exactly the band asymmetry read as an error budget. The figure grows as the fill weakens: at roughly half the standard concentration the corrected Q band falls to 0.036 A and the same error costs 2.8 %, a seventeen-fold asymmetry (`SPEC_capture_quality.md` §16.24.2). Weak fills do not merely add noise; they multiply the leverage of every error that reaches the denominator. Everything at the blue end is a rounding error, and the precision of the whole construction is set by one number: the fitted line's height at roughly 570 nm.

This overturns the intuition that the geometry suggests. The Soret window sits **outside** both anchors — it is an extrapolation some 80 nm beyond the nearest one, which looks like the exposed end and is the first thing a reader worries about. It is not: the baseline lands near zero there, so the long lever costs about one per cent. The fragile band is the **interpolated** one, sitting between the anchors where the geometry is comfortable. The quantity to reason about is not how far a window is from its anchors, but what fraction of that window's absorbance the baseline removes.

5.3b The objection this raises, and what the archive says

If the denominator is that exposed, and if the red anchor is the noisier of the two — it is a third the width of the near window, sits where the lamp is dimmest, and rides the Qy flank — then the metric ought to be hostage to it. On one archived set the far anchor scatters by 0.031 A between re-seats of a single fill, while the entire corrected Q band is 0.068 A. Roughly two fifths of that scatter feeds the baseline at the Q band, so a denominator built this way should carry something like 0.013 A of noise. The measured figure is **0.0019 A**, some seven times smaller.

The reason is that the far anchor and the Q band do not wander independently. Across the runs of a set they move together, and what the subtraction removes is largely the part they share. Measured as a correlation between the two band means, run by run within a set, the figure runs from 0.84 to 0.99, and the corrected band's scatter comes out four to eighteen times below what independent noise would predict.

That result needs one qualification, because part of that agreement is trivial. The fill also changes slowly during a set, and both bands follow concentration, which would correlate them whatever else were true. Removing that axis and correlating what remains separates the two cases. In one set the agreement survives almost untouched — concentration accounts for three per cent of the far anchor's variance, and the two bands still track at 0.99, which is common-mode rejection in the strict sense. In another, concentration accounts for most of the variance and the remaining correlation falls to 0.34, so there the subtraction is removing a concentration drift that the ratio would have divided out anyway. Both mechanisms are present, in proportions that differ from fill to fill.

What survives the qualification is the measured consequence rather than a single explanation: the corrected Q band's scatter really is far below the independent prediction in every set examined. What varies is how much credit the baseline deserves for it, and this is one session's worth of evidence across five sets. It is recorded here because it answers the objection that §3.5 otherwise leaves open — an anchor placed on signal, and on the noisiest part of the spectrum, would seem to be the worst possible choice for the denominator, and in practice it is not.

5.3c Why that is not a coincidence — the two windows share one axis and differ on the other

There is a physical reason the arrangement behaves this way, and it is worth stating because it turns a lucky empirical result into a designed one. The two windows sample the same Q manifold of the same molecule, and that manifold can be moved along two independent axes.

The first is simply **how much pigment is in the beam** — concentration, path length, how the jar happens to sit. Beer–Lambert scales every band of a fixed pigment population by the same factor, so this axis moves the far window and the Q window *together*. It is the axis that varies between re-seats of one fill, and it is precisely what the correlation of §5.3b measures. That the correlation is *positive* is the diagnostic: if speciation were driving the run-to-run scatter, intensity would be moving *between* the two windows and they would vary in opposition.

The second is **speciation** — the loss of magnesium that turns protochlorophyll into protopheophytin. As §3.3 describes, that lowers the symmetry from $D_{4\mathrm{h}}$ to $D_{2\mathrm{h}}$, lifts the degeneracy of the Q states and redistributes intensity out of the reddest band toward the blue. This axis moves intensity *from* one window *into* the other, and leaves the total roughly where it was.

The two axes are cleanly separable in the archive, and the separation is stark:

between green and brown	green	brown	Cohen's *d*
far window alone	0.1566	0.1526	**0.08**
Q window alone	0.1792	0.2251	−0.95
Soret window alone	1.0353	1.0855	−0.35
far ÷ Q — the split between them	0.8673	0.6750	**3.43**

No single window separates the classes. The far window on its own is worthless for it, at *d* = 0.08. What separates them is how the intensity is *divided* between the two, which is exactly what a redistribution mechanism predicts and what a change in amount cannot produce.

This is what makes the far anchor defensible despite §16.10.2a's second violation. It shares the amount axis with the band it is subtracted from, so the fluctuations they have in common — seating, path, concentration — partly cancel in the subtraction. And it differs from that band on the speciation axis, so the quantity the metric exists to detect survives the same subtraction. An anchor genuinely free of pigment, as Morton–Stubbs requires, would share neither: it would cancel less noise and carry no signal. The window is doing two jobs at once because the physics of the two axes allows it to.

⚠ The argument is a reading of the measurements above rather than an independent prediction, and the within-set evidence behind it is one session. It is offered as the reason the arrangement is not accidental, not as proof that it must hold on another instrument.

5.4 ⭐ Why it discriminates — the denominator inverts

	green	brown	green ÷ brown	*d*	separation
A_{Q} *(before)*	0.2300	0.2251	1.02	0.26	overlap
B_{Q} *(after)*	**0.0731**	**0.1008**	**0.73**	**−10.83**	**clean, inverted**
B_{Soret}	1.1323	1.0237	1.11	2.70	clean
Pigment Index	**15.499**	**10.160**	**1.53**	**10.20**	**clean**

Before correction the two classes' Q bands are indistinguishable. After correction the ordering **reverses** and becomes decisive: brown's corrected Q band is 38 % *higher*, at *d* = −10.83 with no overlap between the two sets of six runs.

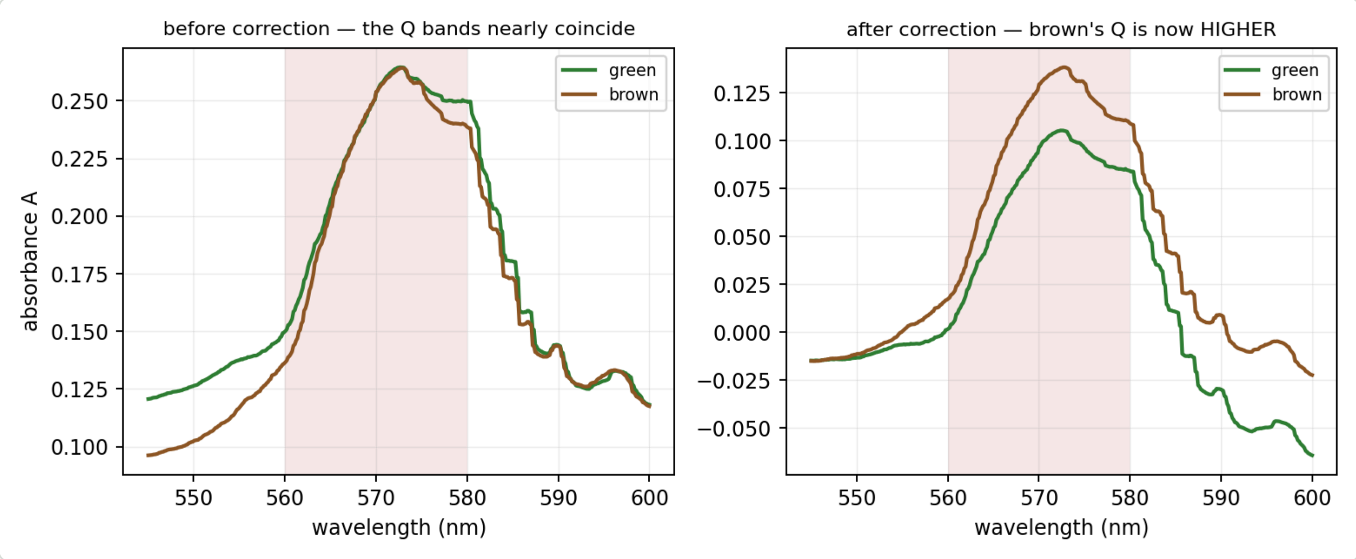


Figure 4 — the Q band before and after correction. Left: the curves lie on top of each other inside the shaded window. Right: after each curve's own baseline is removed, brown stands clearly above green. Nothing about the oils changed between the panels — only what the band is measured *above*.

Why. The baseline under the Q band differs by class — **0.1569** green against **0.1244** brown — so green has *more* subtracted. And green's baseline sits higher because A_{far} is 0.2035 against 0.1526: 620–630 *is* intact pigment (§3.5). The causal chain:

$$\textit{intact pigment raises } A_{\text{far}} \Rightarrow \textit{raises the fitted baseline} \Rightarrow \textit{lowers } B_{\text{Q}} \Rightarrow \textit{raises the index}$$

The two effects multiply:

$$\frac{\text{index}_{\text{green}}}{\text{index}_{\text{brown}}} = \frac{B_{\text{Soret,green}}}{B_{\text{Soret,brown}}} \times \frac{B_{\text{Q,brown}}}{B_{\text{Q,green}}} = 1.106 \times 1.379 = 1.525$$

The numerator contributes ×1.11 and the **denominator ×1.38** — the inverted Q band is by far the larger term, and moving the anchor onto the Qy band nearly doubled its lead (it was ×1.20 on 600–630). A *d* = 2.70 numerator and a *d* = −10.83 denominator compose into *d* = 10.20.

> ⚠ **Why the composed *d* is not simply bigger.** Both components improved, and the class gap grew from 31.7 % to 52.5 % of the brown mean — yet *d* slipped from 11.04 to 10.20 on this pair, because the **scatter grew too**: green's run-to-run CV went 2.89 % → 4.61 %. A 10 nm window collects a third as many points as a 30 nm one and sits in the lamp's dimmest stretch, so it is noisier. On the spec's pooled-green basis the comparison comes out the other way (*d* 9.80 → **10.35**, `SPEC_capture_quality.md` §16.20.4) because pooling two fills adds fill-to-fill scatter, which cost the old anchor more than the new one. **Both are measured; neither transfers to the other's data set.** See §5.8.

This is why the uncorrected ratio fails (Appendix D.1): it divides a signal-bearing Soret band by a Q band that carries nothing. Only after the baseline is removed does the denominator become a discriminator — and then the dominant one.

5.5 The three-region identity

Because the baseline is a straight line read at the band centroids, the metric can be written without mentioning a baseline at all. With t_X the position of band X between the anchor centroids:

$$t_X = \frac{\bar{\lambda}_X - \bar{\lambda}_{near}}{\bar{\lambda}_{far} - \bar{\lambda}_{near}} \qquad t_{Soret} = -0.841 \qquad t_Q = +0.422$$

$$B_X \approx A_X - [(1 - t_X)\,A_{near} + t_X\,A_{far}]$$

$$\text{Pigment Index} \approx \frac{A_{Soret} - 1.841\,A_{near} + 0.841\,A_{far}}{A_Q - 0.578\,A_{near} - 0.422\,A_{far}}$$

Verified against the shipped code: max error 0.02 % (green), 0.05 % (brown).

Read the signs. A_{far} enters the numerator **positively** and the denominator **negatively** — both *raise* the index. This is not "Soret over Q with a correction"; it is a **three-region measurement** in which the red window participates twice, with the largest coefficient of any numerator term save the Soret band itself.

Why the identity is now all but exact. The real fit is a least-squares line through ~206 points and is **steeper** than the naive two-centroid chord — but only by **+1.2 % (green) and +1.6 % (brown)**, against +13.8 % / +5.4 % on the 600–630 anchor. The excess steepness comes from the anchor windows' *internal* slope, and a 10 nm window climbing the Qy flank has far less of it to contribute than a 30 nm one. With the chord and the fit nearly coincident, the identity's error falls to **0.02 %**.

⚠ This improvement arrives **despite** the geometry getting worse, not because of it. The windows' weighted centroid moved from 572.5 nm to **577.5 nm** while the Q band stays at 570.0 — so the extra steepness now pivots **7.4 nm** from the Q band rather than 2.5 nm, and no longer cancels there. It simply has almost nothing left to pivot *with*. The old anchor's identity held by a lucky cancellation; this one holds because the two lines are nearly the same line.

5.6 ⭐ Dilution invariance — the proof

Model the curve as pigment plus a scattering pedestal that does **not** track concentration:

$$A(\lambda) = \varepsilon(\lambda)\,c\,l + P(\lambda)$$

ε extinction, c concentration, l path length, P the turbidity pedestal.

Take $P = 0$ for a moment. Every step from the curve to the two corrected band means — the band mean, the least-squares fit, the subtraction — is **linear and homogeneous** in the data. So scaling the input scales the output identically:

$$B_X = c\,l\,(\bar{\varepsilon}_X - \ell_\varepsilon(\bar{\lambda}_X)) \equiv c\,l\,e_X$$

$$\text{Pigment Index} = \frac{c\,l\,e_{\text{Soret}}}{c\,l\,e_Q} = \frac{e_{\text{Soret}}}{e_Q}$$

c and l are gone. A degree-1 homogeneous functional over a degree-1 homogeneous functional is degree-0. **This is invariance by construction, not by approximation.**

One step is exact and is easily mistaken for an approximation. $B_X = A_X - L(\bar{\lambda}_X)$ holds *exactly*, because the mean of a straight line over a window equals that line at the window's centroid. The only approximation in §5.5 was the slope.

The proof never mentions chlorophyll. It uses only that a component scales with c — so §3.5's far-anchor pigment contamination is **harmless to invariance**. It changes what the metric *means* chemically; it does not change whether it is invariant.

5.7 What breaks invariance — pedestal curvature, and nothing else

Restore P. The fit is linear, so it splits:

$$B_X = c\,l\,e_X + r_X, \qquad r_X = \bar{P}_X - \ell_P(\bar{\lambda}_X)$$

r_X = the pedestal's departure from its OWN best-fit line, at band X.

pedestal	invariance
$P = 0$	exact
P exactly **linear** in λ	**also exact** — the fit removes it completely, $r \equiv 0$
P curved	broken, in proportion to r

So invariance fails only to the extent that turbidity is non-linear across 440–630 nm. Scattering goes roughly as λ^{-n} and a line approximates a curve; that curvature is the *entire* residual error.

The sensitivity, and the denominator is hit ten times harder again:

$$\frac{d\,\ln(\text{index})}{d\,\ln c} \approx \frac{r_Q}{B_Q} - \frac{r_{\text{Soret}}}{B_{\text{Soret}}} \approx \frac{r_Q}{B_Q}$$

The second term drops because B_{Soret}/B_Q is **15.5** (green) and **10.2** (brown).

And the error is concentration-dependent, since r_Q is fixed while B_Q grows with c:

$$\text{sensitivity} \approx \frac{r_Q}{c\, l\, e_Q} \propto \frac{1}{c}$$

Two consequences: a log–log plot of the index against concentration must be **curved**, flattening toward high *c* (a constant slope would falsify the pedestal explanation); and the total error is **U-shaped**, because curvature degrades invariance at *low c* while Soret stray-light compression degrades it at *high c* (`SPEC_capture_quality.md` §16.11.8). An optimal working concentration exists between them.

Against the record: the baseline **halves** the dilution error (5.49 % → 2.75 %). And the residual dependence is bounded directly: pooling the three within-oil dilution pairs on record gives a log–log slope of **+0.033 ± 0.029** — consistent with zero *(`SPEC_capture_quality.md` §16.10.8)*. In practice a realistic preparation error of ±17 % moves the index by about **0.6 %**, and even a *fourfold* dilution error moves it under 5 %, against a class gap of **52 %**. ⚠ That slope is an **upper** bound: each pair is a different fill as well as a different dilution, so it carries fill-to-fill scatter with it.

> **Those two figures were measured on the 600–630 anchor** and are quoted here because they are what the record holds — the halving experiment has not been repeated on 620–630. What *has* been measured on both is the cleanest single dilution pair, post-rebuild: log–log slope **−0.12** on the old anchor against **−0.05** on the new one (§5a.1). The anchor move more than halved the residual concentration dependence — the one number in this chapter that improves for a reason the theory predicts, since a narrower window further from the Soret band gives the pedestal's curvature less room to differ across it.

5.8 Performance

	value
green mean / σ *(n = 6)*	15.499 / 0.714
brown mean / σ *(n = 6)*	**10.160 / 0.197**
gap	**5.339 = 52.5 %** of the brown mean
pooled σ	0.524
Cohen's *d*	**10.20**
green margin to T = 12.5	**+4.20 σ**
brown margin to T = 12.5	**+11.87 σ**

The threshold is T = 12.5 and it was DERIVED, not inherited — the midpoint of the empty corridor between the highest brown run and the lowest green run, rounded (`SPEC_capture_quality.md` §16.20.4). There was no predecessor on this scale to inherit from. ⚠ **T = 10.6 belongs to a different metric** — the pedestal-corrected index of §5a.3 — and to the old 600–630 gauge before it. Reading this chapter's numbers against 10.6 is the single easiest mistake to make with this document.

> ⚠ **Why this green margin differs from the spec's.** `SPEC_capture_quality.md` §16.20.4 scores green as sets **B+C** pooled (n = 12); this document declares green as set **C alone** (n = 6, 5 df). Fewer degrees of freedom, and the t-distribution charges heavily for that. Both are correct for their own data set; **neither may be quoted with the other's.**

> ⚠⚠ **The comparison that must not be made.** The 600–630 anchor scored *d* = **11.04** on exactly this pair, against **10.20** here. That is not a regression in the metric: the class gap grew from 31.7 % to 52.5 % while the noise grew alongside it, and *d* is their quotient. On the spec's pooled basis the same anchor move reads *d* 9.80 → **10.35**, i.e. an improvement. **A single *d* is not a property of a metric — it is a property of a metric and a data set together.** What the anchor move buys is stated in §5a.1 and it is not primarily discrimination; it is dilution invariance, fill-to-fill repeatability, and a baseline anchor that no longer straddles a lamp line.

How well is σ known? With n = 6 the point estimate is not what to plan on. A χ^2 interval on brown's σ gives [0.123, **0.483**] — and **even at the upper bound brown clears the threshold by 4.84 σ**. The conclusion does not depend on six points estimating σ well.

The brown mean also survived a rig rebuild and a different oil: archived `20260727C` reads **10.268** on this anchor from six *fills* on the old rig; series D reads **10.160**, a difference of **−1.05 %**. ⚠ Its scatter does not survive: σ = 0.917 against series D's 0.197, i.e. **4.7×**. That is the rebuild, not the anchor — §16.11 measures the same factor on the old metric — but it is worth seeing that the narrower window did nothing to rescue a badly-seated rig.

(t-distribution throughout, per `SPEC_capture_quality.md` §16.10.11a — the error is heavy-tailed, so the Gaussian is optimistic exactly where it matters.)

5.9 ✅ The 607 nm artifact no longer touches the metric

The lamp's 607 nm emission line used to sit **inside** the far anchor, because that anchor began at 600 nm. On the 620–630 window it does not: **no band this metric reads contains it.**

window	range	contains the 607 nm line?
Soret	440–460	no
Q	560–580	no
near anchor	520–540	no
far anchor	**620–630**	**no** *(600–630 did)*

On the old anchor its contribution was measurable: bridging the line by interpolation across 604–612 nm lowered A_{far} by **7.2 %** (green) and **8.4 %** (brown), moving the index −5.9 % / −4.9 % and *d* from 11.04 to 11.65 — slightly *better* without it. It was never creating the discrimination, but it **was** baked into the absolute scale on which that anchor's threshold had been calibrated. It no longer is.

⚠ **This is a reason the thresholds could not be carried across.** A scale that included a lamp line and a scale that excludes it are not the same scale, whatever else changed with them. It is also why the line's physics still matters below: the lamp is the same lamp, and a *different* lamp would still move things — just no longer through this particular door.

What the artifact actually is

Both named artifacts are **lamp emission lines**, present in the reference spectrum itself:

	position in the reference	height above the local continuum
473 nm feature	~475 nm	**+48 %**

	position in the reference	height above the local continuum
607 nm feature	~606 nm	**+20 %**

A sharp line in the lamp should *cancel* in $T = S/R$, since it appears in both captures. That it does **not** cancel — it survives into the absorbance as a spike — means the two captures do not see it identically. The project's term for this is **registration**: the reference and the sample are two separate jar insertions, and anything that displaces the beam between them (the jar seating, §5.1) shifts where a given wavelength lands on the sensor. A sharp line offset by a fraction of its own width turns into a derivative-shaped spike when the two spectra are divided. The artifact's 2.7 nm width is consistent with a shift of that order.

⚠ **This attribution is the project's, and I could not confirm it cleanly.** Estimating the line's position separately in reference and sample gives an apparent offset, but the estimate is confounded: the oil's own absorbance changes across the window, which tilts the local continuum and drags any centroid measure with it. **The lamp-line origin is verified; the registration mechanism is plausible and untested.** A competing explanation — detector nonlinearity or in-instrument stray light at a bright, high-contrast line — has not been excluded.

⇒ **What would move the threshold** is therefore a change to **the lamp's line spectrum** (a different lamp, or an ageing one) or to **the reference↔sample alignment** (calibration, jar seating, optics). A different *camera* is at most a second-order influence, through how the sensor grid samples the line — an earlier draft of this document overstated it.

5a ⭐⭐ Where the far anchor came from — and the three verdicts *(2026-08-03; `SPEC_capture_quality.md` §16.20)*

Everything above is written on the shipped **620–630 nm** far anchor. It was **600–630** until 2026-08-03, and every number in chapters 4–5 changed when it moved — not because the oils or the instrument changed, but because the line the bands are measured above is drawn through a different window. §5's algebra is unchanged in *form*; only where the second window sits.

This chapter is the record of that move: why it was made, what it cost, and why the bench now shows the index three ways rather than one.

5a.1 Why the window moved — and why it is the direction nobody tried

§5.9 and §5.5 together make the problem plain: the far anchor **straddles** two things it should not. It contains the **607 nm lamp line**, and it stands on the **pigment's own Qy band** — protochlorophyll's Qy is at ~623–626 nm, not chlorophyll's ~665, so that band sits *inside* the window rather than beyond it.

Two earlier attempts moved the wrong way. Pulling the right edge **in** (600→620) collapses the class gap; excising 618–630 as "the lamp's red cliff" deletes the Qy flank and makes the residual *worse*. **Both remove the pigment information.** The move that works keeps it and drops 600–615 instead:

```
old  600 |==============================| 630
         straddles the 607 nm line AND the Qy band

new                     620 |==========| 630
         starts clear of the line, centred on Qy
```

Measured on post-rebuild data, against the shipped anchor:

	600–630 *(old)*	**620–630** *(new)*
Cohen's *d*, green vs brown *(pooled green B+C)*	9.80	**10.35**
Cohen's *d*, green vs brown *(this document's pair)*	**11.04**	10.20
dilution slope `s`	−0.12	**−0.05**
pedestal residual `r_Q`	−0.0246	**−0.0184**
re-seat σ / class gap	12.0 %	**11.4 %**
σ_fill / class gap	6.2 %	**4.4 %**
`M∞` — the scale	9.998	12.450
three-region identity error (§5.5)	0.52 %	**0.02 %**
607 nm lamp line inside the anchor	**yes**	**no**

⚠ **Read the first two rows together.** Discrimination improves on the pooled-green basis and slips slightly on this document's declared pair. That is not a contradiction: *d* is a gap divided by a scatter, and which data set supplies the scatter decides the answer. **The honest summary is that the anchor move is roughly neutral for discrimination** and buys its keep elsewhere — halving the dilution slope, cutting fill-to-fill scatter by a third, and removing a lamp line from the measurement.

⚠ **The scale moves, so `T` must be re-derived** — that cost is real and is the reason this is not simply an improvement. And the gains are **not** what a first look suggested: a sweep scored on *pre-rebuild* fills showed the class gap doubling, and that gain **did not survive** re-scoring on clean data. What survives is the table above.

5a.2 ⚠ `r_Q` belongs to its anchor

`r_Q` is defined as the pedestal's departure from **its own best-fit line** (§5.7). Move the anchor and the line moves, so the residual moves with it:

$$r_Q(600\text{–}630) = -0.0246\ A \qquad r_Q(620\text{–}630) = -0.0184\ A$$

Pairing 620–630 band means with the 600–630 constant is a category error, and it is an easy one to make because both are called `r_Q`. The shipped constant is per-anchor and per-rig-state (§5.7's residual does not survive a mechanical rebuild).

5a.3 The three verdicts, and why the raw ratio lost its

	metric	`T`	why
1	$B_{Soret} / (B_Q - r_Q)$	**10.6**	the pedestal-corrected index — the primary
2	B_{Soret} / B_Q	**12.5**	the same anchor, uncorrected — **this document's Pigment Index**
3	A_{Soret} / A_Q	⛔ **none**	the raw ratio — **shown as a value, with no verdict**

Each adjacent pair isolates one step of the construction. ⚠ The three sit on **different scales**: compare verdicts, never numbers.

The raw ratio carries no threshold because none exists. On post-rebuild data green reads 5.387 ± 0.510 and brown 4.842 ± 0.290 — Cohen's *d* **1.20**, and the classes **overlap outright** (lowest green run 4.863 below the highest brown run 5.340). No line separates them.

⚠⚠ **And its shipped threshold had been wrong for weeks.** `T` = 4.4 sat **below the entire brown class** (minimum 4.622), so every run of the brown reference oil was reported *"good — green"* — on the PUBLISHING badge, the one screen an end user sees. §5.4's argument for why the baselined index discriminates and the raw one does not was right all along; the gauge had simply never been re-scored against a post-rebuild brown series.

5a.4 ⭐⭐ The far anchor's peak, confirmed on two lamps *(2026-08-09; full record in `SPEC_capture_quality.md` §16.28)*

§5a.1 placed the window on 620–630 nm because protochlorophyll's Qy *should* be centred there. That was an argument; it has now been measured. Two runs on the same rig with **two different lamps** — a Sansi V2 and the Yuji — put the lamps' own sharp emission structure 3.4 nm apart, and the absorbance maximum stayed at **629–630 nm** under both. Instrument structure would have moved with the lamp. This did not: the band is real, and the far anchor ends on its peak.

⇒ **That is the session's contribution to this document: confidence in the shipped index went up, not its definition.** `M448` was carried across the same lamp swap and agreed to **3 %** where the uncorrected ratio moved 51 % (⚠ subject to §16.28's open question of whether the two runs shared a fill). Nothing here changes a window, a constant or a threshold.

6. Colour

The evaluation tab carries ten colour chips. They are a presentation feature, and this chapter exists partly to say why they are **not** a verdict input.

6.1 How a spectrum becomes a colour

$$\mathrm{SPD} \Rightarrow \mathrm{CIE}\ XYZ \Rightarrow xy \Rightarrow \mathrm{RGB} \Rightarrow \mathrm{HSL} \Rightarrow hue$$

The spectrum is rebinned to 380–780 nm at 1 nm, integrated against the **CIE 1931 2° colour-matching functions** under **D65**, reduced to chromaticity xy, converted to RGB and read as HSL (`SPEC_spectrum_processing.md` §4). **Luminance is discarded at the $XYZ \to xy$ step**, so every colour reported is chromaticity-derived.

Two converters are used deliberately: absorbance-derived chips use **sRGB** (full gamut), transmission-derived chips use the tuned **rgbxy** module (verdict-compatible, so the pumpkin hue thresholds are untouched). Absorbance values are sanitised first — non-finite → 0, **negatives → 0** (absorbance goes negative where $T > 1$, and negatives corrupt the CIE integral) — and capped by a **relative** ceiling at twice the 99th-percentile value, so a $T \to 0$ spike cannot dominate.

6.2 The three families

family	source	behaviour under dilution
Intrinsic	absorbance	**invariant** — $A \to kA$ leaves chromaticity unchanged
Intrinsic-perceived	absorbance, complemented	invariant
Perceived	transmission	**dilution-dependent** — $T \to T^k$

Intrinsic-perceived is the *colorimetric* complement: reflect the absorbed chromaticity through the D65 white point, $2 \cdot white - absorbed$ — the "other half of the light". This lands ~4° from the true perceived hue, against ~34° for a naive +180° HSL flip.

Each family is shown at its measured saturation and lightness plus a **hue-normalised** twin at fixed S = 38 %, L = 34 %, so that only hue varies between oils. A guard greys any chip whose **chroma** $(1 - |2L - 1|) \cdot S$ falls below 8 % — near-white and near-black have no meaningful hue, and HSL saturation alone would report a false one.

6.3 ⛔ Measured: colour does not discriminate these oils

chip	green *(n = 12)*	brown *(n = 6)*
Intrinsic	H 298–300°	H 298–300°
Intrinsic-perceived	H 67–69°	H 68–69°
Perceived	H 70°	H 70–71°
hue-normalised variants	H 300° · S 38 % · L 34 %	**identical**

This **confirms** `SPEC_capture_quality.md` §16.10.15 ("colour channels do NOT discriminate this oil pair") on post-rebuild data, and extends it from raw channels to the full HSL retrieval path.

Why it fails is instructive. Colour is a *broadband integral* — the CMFs weight the whole visible range — so the narrow, structured differences that carry the class signal (§3.6: a redistribution inside a 50 nm slice of the Q region) are averaged away against the enormous Soret absorbance that both oils share. The metric wins precisely because it looks at **narrow windows**; colour loses for the same reason.

⇒ The chips are worth showing and must never be thresholded.

7. What these numbers do not mean

7.1 A band mean is not dilution-invariant

A_{Soret} separates our two sets at *d* = 2.31 and B_{Soret} at *d* = 2.70, with no overlap. **Do not use them.** The two sets happen to share a dilution recipe, so the comparison is valid only *within* that accident. Only ratios divide the common factor out — which is why the UI marks ratios in bold as decision metrics and shows the band means without thresholds.

7.2 A ratio is invariant only if BOTH sides are corrected the same way

> **A ratio is dilution-invariant if and only if its numerator and denominator are both pedestal-free and corrected the same way.**

The legacy $G = D_{\mathrm{Q}}/A_{\mathrm{clarity}}$ and $G' = D_{\mathrm{Q}}/A_{\mathrm{blue}}$ apply the correction to **one side only** — a locally baseline-corrected numerator over an uncorrected denominator. They are not merely weak discriminators; they are *structurally* incapable of dilution invariance (`SPEC_capture_quality.md` §16.14.3).

7.3 The three verdicts are not on the same scale

The tab shows **three** readings of the same capture (§5a.3), each on its own scale:

	metric	threshold
1	$B_{\mathrm{Soret}}/(B_{\mathrm{Q}} - r_{\mathrm{Q}})$ — baseline **and** pedestal	**10.6**
2	$B_{\mathrm{Soret}}/B_{\mathrm{Q}}$ — baseline only *(this document's Pigment Index)*	**12.5**
3	$A_{\mathrm{Soret}}/A_{\mathrm{Q}}$ — raw	**none**

Only the verdicts are comparable — never the numbers. They are ratios of different quantities, and the two thresholds are not interchangeable: 10.6 belongs to metric 1 and 12.5 to metric 2. ⚠ That 10.6 was *also* the old 600–630 gauge's threshold is a coincidence of arithmetic, not a shared scale — it was retained on metric 1 by explicit decision after being checked against the new corridor, not carried over (`SPEC_roast_ampel.md` §2b).

7.4 ⚠ We do not know what the 560–580 band is

Two candidates, and they are not equivalent: the **Q(1,0)** vibronic satellite of the intact pigment, or a **Qx** band of protopheophytin. Since 560–580 is the index's *denominator*, the difference is not academic — if it is the degradation product, the metric is a literal *intact ÷ degraded* ratio.

Evidence favours the second (§3.6: `A_Q` equal across classes while the 572 feature is *stronger* in brown), but **no source we hold assigns this band**, and the comparison is between two bottles rather than one oil before and after demetallation. Open: `SPEC_pumpkin_peak_ratio_eval.md` §15.5.

7.5 Precision is not correctness, and these are re-seat numbers

Both restated from §1.4 because they are the two most commonly dropped caveats. **T = 12.5 is unvalidated**; and σ_fill — the scatter a real single measurement is subject to — is unmeasured for brown.

8. Reference sheet

Green = `20270729C`, brown = `20260731A`, both the mean of six runs.

shown as	symbol	formula	green	brown	*d*
Soret · 440–460	A_{Soret}	mean A_{d} over 440–460	1.186	1.086	2.31
Q · 560–580	A_{Q}	mean A_{d} over 560–580	0.230	0.225	0.26
Clarity · 510–540	A_{clarity}	mean A_{d} over 510–540	0.114	0.095	1.62
far · 620–630	A_{far}	mean A_{d} over 620–630	0.204	0.153	2.08
Pigment ratio *(verdict 3)*	—	$A_{\mathrm{Soret}} / A_{\mathrm{Q}}$	5.172	4.842	1.18
Pigment ratio · clarity	—	$A_{\mathrm{Soret}} / A_{\mathrm{clarity}}$	10.409	11.572	−1.08
Soret · linear baseline	B_{Soret}	mean B over 440–460	1.132	1.024	2.70
Q · linear baseline	B_{Q}	mean B over 560–580	0.073	0.101	−10.83
Pigment ratio · linear baseline *(verdict 2)*	**Pigment Index**	$B_{\mathrm{Soret}} / B_{\mathrm{Q}}$	**15.499**	**10.160**	**10.20**
· with the pedestal put back *(verdict 1)*	—	$B_{\mathrm{Soret}} / (B_{\mathrm{Q}} - r_{\mathrm{Q}})$	**12.380**	**8.590**	**9.33**
Greenness G	G	$D_{\mathrm{Q}} / A_{\mathrm{clarity}}$	—	—	−1.99
Pigment D_Q	D_{Q}	peak above a local line	—	—	−2.48
Soret A_blue	A_{blue}	reference-gated 450–490	—	—	1.18
Pigment ratio · legacy	—	$A_{\mathrm{blue}} / A_{\mathrm{clarity}}$	—	—	0.11
G' (alt.)	G'	$D_{\mathrm{Q}} / A_{\mathrm{blue}}$	—	—	−5.33

Constants

constant	value	where
capture window	440–630 nm	`WAVELENGTH_MIN_NM` / `MAX`
frames per capture	150	`FRAMES`
de-spike kernel	7 points ≈ 1 nm	`MedianFilterOp`
reference floor	6.31×10^{-5} of peak	`TransmissionLogicModule`
division guard ε	10^{-3}	`DevSpectralPlugin`
hue-normalised S / L	38 % / 34 %	`DevSpectralPlugin`
achromatic chroma guard	8 %	`EvaluationColorUtil`
far anchor	**520–540 + 620–630 nm**	`PB_BASELINE_WINDOWS`
pedestal residual r_{Q}	**−0.0184 A** *(this anchor's own, §5a.2)*	`PB_R_Q`
verdict threshold T	**12.5** on the Pigment Index	`SPEC_capture_quality.md` §16.20.4

constant	value	where
verdict threshold T	10.6 on the pedestal-corrected index	`SPEC_roast_ampel.md` §2b

Appendix A — the least-squares fit

§5.2 fits a straight line through the two anchor windows. This is what "fit" means.

A.1 Residuals, and why they are squared

A trial line $A = m\lambda + c$ will not pass exactly through every measured point. The **residual** at each wavelength is how far the measurement sits above or below it:

$$r(\lambda) = A_{\mathrm{d}}(\lambda) - (m\lambda + c)$$

positive where the curve lies above the trial line, negative where it lies below.

"As close as possible to all the points at once" is then made precise by minimising the **sum of the squared** residuals. Squaring does two things: it removes the sign, so points above and below the line cannot silently cancel each other out; and it penalises one large miss more heavily than several small ones, which is what makes the fit follow the bulk of the data rather than being dragged by outliers on one side.

A.2 The weighting, and why it is not one-point-one-vote

$$\mathrm{SSR}(m, c) = w_{\mathrm{near}} \Sigma_{\lambda \in \mathrm{W}_{\mathrm{near}}} r(\lambda)^2 \quad + \quad w_{\mathrm{far}} \Sigma_{\lambda \in \mathrm{W}_{\mathrm{far}}} r(\lambda)^2$$

the weighted sum of squared residuals, one term per anchor window.

with per-point weights $w_{\mathrm{near}} = 1/|W_{\mathrm{near}}|$ and $w_{\mathrm{far}} = 1/|W_{\mathrm{far}}|$ — each window's points weighted by one over that window's point count, so **each window contributes total weight 1**.

> **Why not weight every point equally?** The far window holds 212 points against the near window's 135, so an unweighted fit would let the red end pull about **1.6× harder** — and, worse, *widening a window would silently move the baseline* as a side effect. A window is one piece of evidence about where the baseline runs, not one piece per sample. Measured across the 25 runs of 2026-07-27 the two fits differ by ~0.5 % and change **no verdict**: this is for predictability under a window change, not accuracy.

A.3 The minimisation

$$(m, c) = \mathrm{argmin}_{m,\, c}\ \mathrm{SSR}(m, c)$$

read: the slope and intercept AT WHICH that sum is smallest.

argmin denotes the *argument* that minimises — the pair (m, c), not the value of the sum itself. Nothing is actually searched for: setting the two partial derivatives to zero gives a pair of linear equations with a closed-form solution. The implementation calls `numpy.polyfit` of degree 1 with `w = sqrt(weights)`, because `polyfit` weights the *residual* rather than its square.

Appendix B — Cohen's *d*: the variants, and which is used where

§1.3 gives one formula. There are several in circulation, they do not always agree, and this appendix says which this project uses and why.

B.1 The family — Cohen's *d* is one member of it

Standardised mean difference (SMD) is the umbrella term: *a mean difference expressed in standard-deviation units*. It does not say **which** standard deviation, and that is exactly where the named variants differ:

variant	denominator	when it is the right choice
Cohen's *d*	the **pooled** SD of both groups	two groups of comparable status — **our case**
Hedges' *g*	the same, × a small-sample bias correction	the same, at small *n* — see B.3
Glass's Δ	the **control** group's SD alone	when one group is a reference standard whose scatter is the meaningful yardstick

So Cohen's *d* is *an* SMD, not a synonym for it — although many fields (Cochrane, for one) say "SMD" as the umbrella and then report a specific recipe. **This project uses Cohen's *d* and writes it *d*.** Glass's Δ would be defensible once we have a certified reference oil; we do not, and neither of our two classes is a control.

B.2 ⚠ Two pooled-SD conventions — and they diverge at unequal *n*

Within Cohen's *d* itself there are two formulas for the denominator:

$$s_{\text{pooled}}^{\text{RMS}} = \sqrt{\frac{s_1^{\,2} + s_2^{\,2}}{2}}$$

the simple root-mean-square of the two standard deviations.

$$s_{\text{pooled}}^{\text{df}} = \sqrt{\frac{(n_1 - 1)\,s_1^{\,2} + (n_2 - 1)\,s_2^{\,2}}{n_1 + n_2 - 2}}$$

the degrees-of-freedom-weighted form: the larger group's scatter counts for more.

When the two groups are the same size these are algebraically identical — substitute $n_1 = n_2 = n$ into the second and it collapses into the first. They part company only when the groups differ in size, and then the df-weighted form is the conventional choice.

Where each applies in this project:

comparison	*n*	conventions agree?	*d*
everything in this document — green `20270729C` vs brown `20260731A`	**6 vs 6**	**✔ identical**	**10.20**
`SPEC_capture_quality.md` §16.20.4 — green sets **B+C pooled** vs brown	**12 vs 6**	**⛔ differ**	df-weighted **10.35**

⚠ **The one unequal-*n* comparison is in the spec, not here**, and there the two conventions differ by 12 %. `diagnostics/brown_series_d.py` computes the RMS form; the **df-weighted 9.80 is the more defensible figure to quote**. Nothing downstream changes — the per-class σ-margins and error rates are computed separately and are untouched — but the number should not be quoted without naming its recipe.

B.3 Hedges' *g* — the small-sample correction

Cohen's *d* is biased **upward** when samples are small, and n = 6 is small. Hedges' correction removes most of that bias:

$$g = d \cdot \left(1 - \frac{3}{4\,df - 1}\right), \qquad df = n_1 + n_2 - 2 = 10$$

$$g = 10.20 \cdot 0.923 = 9.42$$

So the bias-corrected separation is ≈ 9.4, not 10.20 — about 8 % lower. It changes no conclusion in this document, both being far beyond "large", but *g* is what a statistician would ask for at this sample size and the figure to quote externally.

> **Convention adopted here:** every *d* reported in this document and in the specs is **uncorrected Cohen's *d* on the RMS pooled SD**. Multiply by **0.923** for Hedges' *g* on the 6-vs-6 comparisons. Where a figure is quoted outside the project, quote *g*.

Appendix C — reproducing the numbers

Every value in this document is recomputed from the archived report PDFs' embedded workflow JSON by diagnostics that call the **shipped** `SpectrumFeatureUtil` and `DevSpectralPlugin` code paths, so the document cannot drift from the application.

script	produces
`diagnostics/metric_walkthrough.py`	every intermediate quantity of the chain, both classes
↳ `METRIC_ANCHOR=600 …`	the same, on the superseded 600–630 anchor — the §5a comparison column
`diagnostics/qband_shape.py`	the speciation-vs-concentration test and the resolution measurement
`diagnostics/brown_series_d.py`	the series-D discrimination statistics
`diagnostics/metric_algebra_plots.py`	figures 2–4
`docs/tools/build_pigment_figures.py`	figure 1

Cohen's *d* is the pooled-SD standardised difference on n = 6 per class (defined in §1.3); with six runs a *d* of this size is bounded well away from zero but its point value is loose.

⭐ **`METRIC_ANCHOR` is the one switch that reproduces this whole document on either anchor.** `metric_walkthrough.py` reads it, and `metric_algebra_plots.py` follows whatever the walkthrough is set to, so text and figures cannot drift apart. Default **620** — the shipped window. Every number in chapters 4–5 and every figure 2–4 comes out of one run at that default; §5a's comparison column comes out of one run at `METRIC_ANCHOR=600`.

Appendix D — the legacy metrics *(historical)*

The **"Metrics (dev)"** tab predates the literature bands and uses older machinery on different windows: `BLUE_BAND` 450–490, `GREEN_BAND` 510–540, `Q_SEARCH` 565–590. It is retained so that old numbers stay comparable, and for no other reason. All *d* values are green `20270729C` vs brown `20260731A`.

D.1 The uncorrected ratios

$$S/Q_{\text{plain}} = \frac{A_{\text{Soret}}}{\max(A_{\text{Q}},\ \varepsilon)} \qquad S/Q_{\text{clarity}} = \frac{A_{\text{Soret}}}{\max(A_{\text{clarity}},\ \varepsilon)}$$

	green	brown	*d*	verdict
S/Q_{plain}	5.172 ± 0.267	4.842 ± 0.291	1.18	**overlaps**
S/Q_{clarity}	10.409 ± 0.601	11.572 ± 1.401	−1.08	**overlaps, and inverted**

Neither works. S/Q_{clarity} is worse than useless — it points the wrong way and its brown scatter is 14 %. §5.4 explains why: the uncorrected Q band carries no class information.

D.2 The Q-band peak height D_{Q}

Rather than a window mean, this finds the local maximum in the search band and measures its height above a **local** two-anchor line:

$$(\lambda_{\text{Q}}, A_{\text{peak}}) = \operatorname{argmax}_{\lambda\,\in\,[565,\,590]} A(\lambda)$$

$$L(\lambda_{\text{Q}}) = \overline{A}_{[550,560]} + (\overline{A}_{[595,605]} - \overline{A}_{[550,560]})\,\frac{\lambda_{\text{Q}} - 555}{600 - 555}$$

$$D_{\text{Q}} = A_{\text{peak}} - L(\lambda_{\text{Q}})$$

The anchors are ±5 nm windows around 555 and 600 nm. A flat offset lifts peak and line equally, so D_{Q} is already immune to an additive pedestal — which is why its de-spiked twin barely moves.

D.3 The reference-gated blue band A_{blue}

$$A_{\text{blue}} = \text{mean}\{A(\lambda) : \lambda \in [450, 490],\ R(\lambda) \geq 0.25 \max_{[450,465]} R,\ A(\lambda) \leq 1.5\}$$

Two gates: wavelengths where the *reference* is weak are dropped (trimming the LED's cyan dip), and saturated wavelengths are dropped. It is the only metric that reads the reference spectrum directly.

D.4 The derived ratios, and how they perform

$$G = \frac{D_Q}{A_{\text{clarity}}} \qquad G' = \frac{D_Q}{A_{\text{blue}}} \qquad S/Q_{\text{legacy}} = \frac{A_{\text{blue}}}{A_{\text{clarity}}}$$

metric	*d*	note
Pigment Index *(§5, for comparison)*	**10.20**	the only usable one
G'	−5.33	**sign inverted** — brown reads higher
D_Q	−2.48	inverted
G "Greenness"	−1.99	inverted — despite the name
A_{blue} "Soret"	1.18	weak
A_{clarity}	0.54	none
S/Q_{legacy}	**0.11**	**useless**

⚠ **Three are inverted with respect to their names.** "Greenness G" reads *higher* on the brown oil. The names are historical; `SPEC_pumpkin_peak_ratio_eval.md` §11 found the direction reversed and the fields were renamed once already ("Browning A_blue" → "Soret A_blue"). **Treat every label on the legacy tab as a hypothesis, not a description.** §7.2 adds that G and G' cannot be dilution-invariant either.

References

Pigment identity and band positions

1. **Fruhwirth, G. O. & Hermetter, A. (2007).** *Seeds and oil of the Styrian oil pumpkin: components and biological activities.* Eur. J. Lipid Sci. Technol. **109**(11), 1128–1140. DOI 10.1002/ejlt.200700105. §3.5 identifies protochlorophyll a/b and protopheophytin a/b; Fig. 3 gives the 635 nm fluorescence maximum. Local copy in `spectracs-references/articles/`; free mirror at gruenesglueck.com.
2. **Protochlorophyllide spectral forms.** Pak. J. Biol. Sci. (2010) — scialert.net. Qy ≈ 623 nm (80 % acetone), ≈ 626 nm (methanol); Soret ≈ 440 nm; minor bands 505, 535, 606 nm.
3. **Gouterman, M. (1961).** *Spectra of porphyrins.* J. Mol. Spectrosc. **6**, 138 — the four-orbital model that names the Soret and Q bands.
4. **Histolocalisation of the oil and pigments in the pumpkin seed** — ResearchGate. Speciation as 2,4-divinyl-protochlorophyll a, 2,4-divinyl-protopheophytin a, 2-vinyl-protopheophytin a; protopheophytins 1.1–35.5 % of protochlorophylls, rising with storage.
5. **The four porphyrin spectral types and their Q-band intensity ordering** — *The Use of Spectrophotometry UV-Vis for the Study of Porphyrins*, InTech; Porphyrin overview, ScienceDirect. Etio IV > III > II > I; rhodo III > IV > II > I; oxo-rhodo and phyllo IV > II > III > I — **band I weakest in all four**.
6. **Chlorophyll *a* comparison values** (the molecule we do *not* have) — PhotochemCAD.

⚠ **Not independently sourced:** a primary measurement of *protopheophytin a*'s band positions. §3.3's direction is argued from the classification in [5], which is textbook porphyrin spectroscopy, but the specific numbers for this molecule are not sourced here.

Colour science

1. **CIE 1931 2° colour-matching functions** and the D65 illuminant, via the `colour-science` package; pipeline documented in `SPEC_spectrum_processing.md` §4.
2. **Kreft, S. & Kreft, M.** — the dichromaticity index, with pumpkin seed oil as its type example.

Owning specifications

topic	document
the baseline's derivation, error budget, series D	`SPEC_capture_quality.md` §16.10–16.14
what the anchors contain; the far-window sweep	`SPEC_capture_quality.md` §16.12.12–14
speciation vs concentration; the resolution argument	`SPEC_capture_quality.md` §16.13.9
dilution-invariance algebra	`SPEC_capture_quality.md` §16.14
the three-region algebra; pre-registration	`SPEC_capability_proof.md` §2.1a, §11.4f
legacy metrics; band naming; the open 560–580 assignment	`SPEC_pumpkin_peak_ratio_eval.md` §11, §15
colour retrieval	`SPEC_color_retrieval.md`, `SPEC_spectrum_processing.md`
pigment chemistry in full	`DOC_sample_physics.md`, `KB_spectroscopy_physics.md` §4.1